ECON10005 Quantitative Methods 1

Tutorial in week 8: Week 7 Material

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Introduction

Zheng Fan

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Don't be shy if you need help

- Discuss on Ed Discussion Board
- Attend consultation sessions: see Canvas for time and location
- Consult Stop 1, in case of special considerations,
- Contact QM-1@unimelb.edu.au for admin issues
- Send me an email: fan.z@unimelb.edu.au

Roadmap

1. Common feedback for group draft
2. Hypothesis testing: general routine
3. Tutorial practise questions

Group Draft – Some Common Feedback

Please carefully address the following points for your final project:

- Choose a more specific and engaging **title** for your project.
- Always include **units** for variables, axes, and tables (e.g., thousands, millions).
- Use clear and meaningful variable names (e.g., annual income), rather than raw names (e.g., `var1`, `an_inc`) in all tables, figures and discussions.
- Ensure all plots are properly formatted: include informative titles and axis labels, and avoid unclear or hidden elements.

Group Draft – Some Common Feedback

Please carefully address the following points for your final project:

- For scatter plots, carefully consider which variable is placed on the x-axis and which on the y-axis.
- Present key tables and figures in the main report, and place technical details or supplementary materials in the appendix for interested readers.

The Five Steps of Hypothesis Testing

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4. **Make a decision:** Reject or fail to reject H_0
5. **Answer the question:** Interpretation

Questions

In an attempt to understand the financial literacy levels of men and women in Australia, a researcher collected a sample of **687** Australians. Each individual was asked to complete a commonly used questionnaire containing twelve financial literacy questions. Data (financial_Lit.xlsx) contains the scores for the questionnaire and the gender reported by the individuals.

Perform hypothesis tests to answer the following questions. Make sure you define the population parameters and follow all five steps of hypothesis testing.

1. Financial literacy levels are believed to have improved over the years. Past research shows that Australians, on **average**, have scored **6.4** (out of 12) for the twelve financial literacy questions. Do we have sufficient evidence at a **5% level of significance** that Australians' financial literacy levels have increased since then? Follow the p-value approach.

Question 1

Population Parameter:

- μ = population mean financial literacy score of all Australians

Step 1: Set up the hypotheses The null and the alternative hypotheses for this test are:

$$H_0 : \mu = 6.4 \text{ (mean score has not changed)}$$
$$H_1 : \mu > 6.4 \text{ (mean score has increased)}$$

Note: This is a upper-tailed test because we're specifically testing if scores have increased.

Question 1

Step 2: Determine the decision rule for the given level of significance ($\alpha = 0.05$)

- Determine critical value or p-value:
- If we were using the critical value approach:
- Upper-tail Critical value: $t_{0.05,687-1} =$

Question 1

Step 2: Determine the decision rule for the given level of significance ($\alpha = 0.05$)

- Determine critical value or p-value:
- If we were using the critical value approach:
- Upper-tail Critical value: $t_{0.05,687-1} = 1.647$ (how? from t table)
- Therefore, the decision rule is: Reject H_0 if the test statistic > 1.647
- Since we are asked to use the p-value approach, our decision rule is:
- Reject H_0 if the $p-$ value < 0.05

Question 1

Step 3: Calculate Test Statistic

Follow the Excel file for details of calculations

	Female	Male	All
Average Scores	5.767	7.534	6.649
sd	3.079	2.980	3.154
Number	344	343	687

Question 1

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Follow the Excel file for details of calculations

	Female	Male	All
Average Scores	5.767	7.534	6.649
sd	3.079	2.980	3.154
Number	344	343	687

test-statistics:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{6.649 - 6.4}{3.154/\sqrt{687}} = \frac{0.249}{0.1203} = 2.0709$$

p-value:

Question 1

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p-value: $P(t > 2.0709) = 0.0194$

Question 1

Step 4: Make a decision

We are asked to use the p-value approach:

- Since the p-value $= 0.0194 < 0.05$, we reject H_0.

Question 1

Step 4: Make a decision

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- Since the p-value $= 0.0194 < 0.05$, we reject H_0.

Step 5: Answer the question

- We reject the null hypothesis H_0 against the alternative H_1
- Conclusion: we have sufficient evidence at a 5% significance level that the average financial literacy score has increased over the years.

Questions

In an attempt to understand the financial literacy levels of men and women in Australia, a researcher collected a sample of **687** Australians. Each individual was asked to complete a commonly used questionnaire containing twelve financial literacy questions. Data (financial_Lit.xlsx) contains the scores for the questionnaire and the gender reported by the individuals.

Perform hypothesis tests to answer the following questions. Make sure you define the population parameters and follow all five steps of hypothesis testing.

2. If you conducted the hypothesis test with a significance level of 10%, what would your decision be? Do you have to run the test again?
3. If you are asked to conduct the test at a 1% significance level, do you have to rerun the test? If so, which steps?

Question 2: Changing Significance Level to 10%

Question: If you conducted the hypothesis test with a significance level of 10%, what would your decision be? Do you have to run the test again?

Analysis:

- Previous p-value: $p \approx 0.0194$
- New significance level: $\alpha = 0.10$
- Comparison: $p < \alpha$ $(0.0194 < 0.10)$

Decision:

- We still reject H_0
- No need to recalculate test statistic or p-value (but need to find the new critical value if use critical value to make decision i.e. $t_{0.1,686}$)
- Only comparing the previously calculated p-value to a different α

Question 3: Changing Significance Level to 1%

Question: If you are asked to conduct the test at a 1% significance level, do you have to rerun the test? If so, which steps?

Analysis:

- Previous p-value: $p \approx 0.0194$
- New significance level: $\alpha = 0.01$
- Comparison: $p > \alpha$ (0.0194 >0.01)

Decision:

- We thus **fail** to reject H_0
- No need to recalculate test statistic or p-value (but need to find the new critical value if use critical value to make decision i.e. $t_{0.01,686}$)
- Only comparing the previously calculated p-value to a different α

Question 4: Has the Proportion of Women Passing Increased?

Past research also shows that 55% of women who did the test containing the twelve financial literacy questions passed it. Do we have sufficient evidence at a 5% significance level that the proportion of women who pass the test has increased? Follow the critical value approach.

Summary:

- Past research: 55% of women passed the financial literacy test
- Current study: Sample of women from the 687 Australians
- Significance level: $\alpha = 0.05$
- Approach: Critical value

Question 4: Step 1 - Define Parameters & Hypotheses

Population Parameter:

- p = true population proportion of women who pass the financial literacy test

Hypotheses:

$$H_0 : p = 0.55 \text{ (proportion has not changed)}$$
$$H_1 : p > 0.55 \text{ (proportion has increased)}$$

Question 4

Step 2: Determine the decision rule for the given level of significance ($\alpha = 0.05$)

- Determine critical value or p-value:
- If we were using the critical value approach:
- Upper-tail Critical value: $t_{0.05,344-1} =$

Question 4

Step 2: Determine the decision rule for the given level of significance ($\alpha = 0.05$)

- Determine critical value or p-value:
- If we were using the critical value approach:
- Upper-tail Critical value: $t_{0.05,344-1} = 1.649$ (how? from t table)
- Therefore, the decision rule is: Reject H_0 if the test statistic > 1.649
- Since we are asked to use the p-value approach, our decision rule is:
- Reject H_0 if the $p-$ value < 0.05

Question 4

Step 3: Calculate Test Statistic

From excel, $\hat{p} = 0.535$ *and* $n = 344$ *women*

$$t = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Question 4

Step 3: Calculate Test Statistic

From excel, $\hat{p} = 0.535$ *and* $n = 344$ *women*

$$t = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

test-statistics:

$$t = \frac{0.535 - 0.55}{\sqrt{\frac{0.535 \times 0.465}{344}}} = -0.5621$$

p-value:

Question 4

Step 3: Calculate Test Statistic

From excel, $\hat{p} = 0.535$ *and* $n = 344$ *women*

$$t = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

test-statistics:

$$t = \frac{0.535 - 0.55}{\sqrt{\frac{0.535 \times 0.465}{344}}} = -0.5621$$

p-value: $P(t > -0.5621) = 0.7128$

Question 4

Step 4: Make a decision

We are asked to use the critical value approach:

- Since the est statistic $-0.5621 <$ critical value $t_{0.05,343} = 1.649$, we do not reject the null hypothesis
- If we use p-value, since p-value $0.7128 > 0.05$, we do not reject the null
- Note: both methods always give the same decision

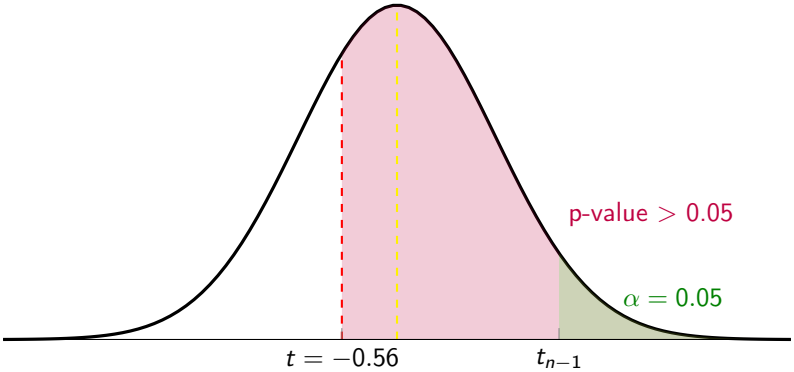
Step 5: Answer the question

- We do not reject the null hypothesis
- Conclusion: our data don't support the claim at a 5% significance level that the proportion of females who passed the financial literacy test has increased over the years.

Question 5

In a sketch of a $t_{(n-1)}$ distribution, illustrate the p-value obtained for the hypothesis test in Question 4.

Question 5



This illustration shows that the p-value is much larger than the significance level, which is expected, given that the test produced a negative test statistic even though we had an upper-tail test.

Key Takeaways

- Hypothesis testing provides a structured framework for statistical inference
- The choice of significance level α impacts the threshold for evidence (eg critical value)
- P-value and critical value approaches lead to the same conclusion

Any final questions?

Thanks for your attention! 😄

Let me know if you have any questions.